

Re: Depth and Stability of Banking System Subindex Evaluation

Dear Professor Allayannis,

Thank you for your thorough and thoughtful evaluation of the Global Capitalism Index and the Depth and Stability of Banking System Subindex. Your critiques are valuable, and we appreciate the care you took in reviewing across multiple elements of the index. We have made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

I. Comments on Banking Depth

Financial Depth vs. Banking Depth

“I would change it in the document from Financial Depth as Financial Depth is much broader than Bank Depth – they could be correlated – but I would think that Financial Depth could include insurance, asset managers etc.”

We agree that increased specificity in component names is beneficial. We have updated the name from “Financial Depth” to “Banking System Depth.”

“I don’t have a lot to say on this subindex; you all have done a good job assembling data and indices that may proxy for Bank depth at a country level. Here are a few comments: I am not sure I understand the first metric “World-Bank 5 Asset concentration”; what is the underlying hypothesis? That concentrated banking systems lack depth? I would disagree with such assessment; a counter example is the US.”

The World Bank 5 Bank Asset Concentration dataset measures the assets of the five largest banks as a share of total commercial banking assets. Our underlying hypothesis is that greater asset depth among the five largest banks generally reflects a deeper and more developed banking system overall. This interpretation is consistent with your example of the United States.

“Several of these metrics are linked to country development. I understand that country development would be likely associated with bank depth. But it might be good to try to somehow scale/control for GDP or something like that. For example, bank branches per 1000 people may just reflect how rich the country is.”

We agree that many indicators of banking system depth correlate to overall levels of economic development. However, the purpose of this subindex is to measure the institutional reach and accessibility of the banking system rather than to isolate banking depth from development itself. Indicators such as bank branches per 100,000 adults, digital payment usage, and credit/debit card penetration capture the degree to which financial services are embedded within the broader economy and accessible to households and firms. Scaling these measures by GDP would risk obscuring the structural differences we seek to measure. A country with a large economy but limited financial intermediation would appear artificially similar to a smaller economy with highly developed financial access if both were normalized by GDP.

By contrast, variables such as private credit to GDP and deposit money bank assets to GDP are intentionally scaled by GDP because they are designed to measure the overall size and intermedi-

ation capacity of the financial sector relative to the economy it serves. Without normalization by GDP, larger economies would appear to have deeper banking systems simply due to scale.

II. Comments on Banking Stability

Regulation

“Again, I would commend you for your efforts to assemble metrics and gather data to reflect bank stability. Here are some comments: I think something to add would be quality of regulation; I don’t unfortunately know of data on this but perhaps this may be linked to country development, GDP/Capita etc. Of course, a counter example to this is the US and Switzerland. One coarse way would be perhaps to assign 0/1 if a country follows Basel III or not. Another is to look at enforcements (again I am not sure whether data exists across countries and over time).”

We agree that the quality of financial regulation is an important determinant of banking system stability, but constructing a globally comparable measure of regulatory quality presents significant data challenges. Many potential proxies, such as GDP per capita, risk conflating regulatory quality with overall economic development rather than capturing regulatory institutions themselves. Your suggestion to code Basel III adoption is a useful example of a rule-based proxy, but variation in adoption and enforcement across jurisdictions make it difficult to construct a consistent cross-country time series. Additionally, this binary indicator would likely not capture enough variation over time to contribute meaningfully to the PCA process.

Similarly, cross-national data on regulatory enforcement actions remain sparse and inconsistently reported across countries. For these reasons, we focus the banking stability subindex on observable balance-sheet and systemic risk indicators that are consistently measured across countries and over time. We will be sure to revisit this as reliable cross-national measures of banking regulatory quality or enforcement become available in the future.

Other Ratios/Metrics

“There are also various other ratios/metrics that can be used for bank stability: For capital, I would use Tier 1/Risk-weighted assets (currently I believe you are using Total Capital to RWA) since Tier 1 is the most loss-absorbing capital for banks. Interestingly, the definition of Tier 1 has dramatically changed once Basel III came into effect (what makes it trickier is that not all countries phased-in Basel III at the same time and/or set rules as to which banks should follow them -commonly the largest).”

We note the suggestion to use Tier 1 capital to risk-weighted assets as a more precise measure of bank capitalization. Our Banking System Stability component already includes the World Bank’s Bank Capital to Assets Ratio, which is defined as Tier 1 capital divided by total assets. The indicator defines Tier 1 capital as the most loss-absorbing, high-quality forms of capital such as common equity, disclosed reserves, retained earnings, and other regulatory-approved instruments that meet regulatory requirements under the Basel framework.

“Regarding a measure of liquidity I would also include the Basel III metric of LCR (Liquidity Coverage Ratio) which equals “HQLA (high quality liquid assets)/Total Net Cash Outflows for 30 days which should exceed 100% to be compliant.”

The LCR is a great addition for the stability component and we now have added it to our subindex, sourced from the IMF’s Financial Soundness Indicators database.

“Another liquidity metric that is commonly used is the ratio of “Loans to Deposits” as well as the simple “Cash to Assets”.”

This is a great suggestion. We include a World Bank dataset measuring “The financial resources provided to the private sector by domestic money banks as a share of GDP” within our banking depth component to measure the extent of bank lending in a given economy. At your recommendation, we added a World Bank dataset measure “The ratio of the value of liquid assets (easily converted to cash) to short-term funding plus total deposits.” This satisfies the cash to assets suggestion.

Too Big to Fail

“A broader issue is that the stability of the banking system does not depend on small banks – there is probably room for many small banks to fail and the system be very stable. You may be controlling this somehow but couldn’t tell. I would focus on the “Too-big-to-fail” banks for each country – perhaps the top 5 or so though I admit this is arbitrary.”

We agree that overall banking system stability is driven primarily by the largest, most important banks rather than the many smaller institutions. To capture the role of dominant banks in shaping banking system depth, we include the World Bank 5-Asset Concentration metric in our Banking System Depth component, which measures the share of total commercial banking assets held by the five largest banks in a country. As discussed on Page 1 under **Financial Depth vs. Banking Depth**, our underlying hypothesis is that greater asset depth among the five largest banks generally reflects a deeper and more developed banking system overall. While this metric reflects the concentration of assets rather than inherent stability, combining it with other stability indicators allows us to reflect both the size and concentration of banking institutions and their financial health.

Credit Quality

“For credit quality additional measures would be NCOs/Total Loans (this ratio uses the actual charge-offs as opposed to NPLs). Another ratio would be the “Coverage ratio” which is “Reserves/NPLs”) – this gives you a sense of what percentage of NPLs are covered by reserves.”

We include the World Bank’s Non-Performing Loans to Gross Loans indicator under our Banking System Stability component as a primary measure of credit quality because it provides broad global coverage and consistent time-series data across countries. We agree that measures such as Net Charge-Offs to Total Loans and the NPL Coverage Ratio (Reserves/NPLs) would provide valuable additional insight into realized losses and the adequacy of loss provisioning. However, these indicators are not consistently available across countries and over time at a global scale.

Banking Profitability Measure

“A measure of profitability of the banking sector would also indicate higher stability (though this too needs to be caveated). Common measures would be ROE and ROA.”

We agree that profitability can serve as an important indicator of banking system stability, as consistently profitable banks are generally better able to absorb losses, build capital buffers, and maintain operations during periods of financial stress. Accordingly, we've added in Return on Assets (ROA) as a component of the Banking System Stability measure. ROA captures the ability of banks to generate earnings from their asset base and is better than ROE for cross-country banking comparisons because it is less sensitive to differences in leverage across banking systems.

JAR 2016 Paper

"Banks' Financial Reporting and Financial System Stability by Viral Acharya"

Thank you for flagging this paper. Acharya's work on bank transparency is compelling, but comparable cross-country time-series data on bank reporting quality do not yet exist. To approximate the stability mechanisms highlighted in this paper, we prioritize observable balance-sheet indicators of liquidity, asset quality, and profitability to capture the financial resilience of banking systems across countries.

Concluding Remarks

Your review raised several important questions regarding the conceptual framing of banking depth, the measurement of stability, and the selection of specific financial indicators. In response, we clarified the scope of the subindex by renaming the component to "Banking System Depth," refined the explanation of the World Bank 5-Asset Concentration measure, and expanded the stability component by incorporating additional indicators such as the Liquidity Coverage Ratio (LCR) and Return on Assets (ROA). We also reviewed several additional measures you suggested and explained where coverage constraints limit their inclusion. We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to strengthen the Depth and Stability of the Banking System subindex and the Global Capitalism Index.

Sincerely,

UVA Democracy and Capitalism Lab